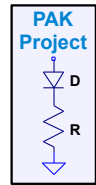


PAK Project – Outline

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Level 1 – How to solve basic diode and transistor circuits

- PAK100 – Preface
- PAK101 – Welcome to the W-function analytical approach
- PAK102 – W-function analytical solution for the D+R
- PAK103 – Basic BJT W-function analytical solution using the D+R solution
- PAK104 – Banwell's CE W-function solution with shunt feedback
- PAK105 – CE W-function solution with shunt feedback *and emitter degeneration*
- PAK106 – General solution for $y \cdot e^y = e^x$ or $y + \ln(y) = x$
- PAK107 – General solution for $y \cdot e^{y^2} = e^x$ or $y^2 + \ln(y) = x$
- PAK108 – General solution for $y^2 \cdot e^y = e^x$ or $y + 2 \cdot \ln(y) = x$
- PAK109 – General solution for $(1/y)e^y = e^x$ or $y - \ln(y) = x$
- PAK110 – General solution for $(1/y^2)e^y = e^x$ or $y - \ln(2y) = x$
- PAK111 – General solution for $y + e^y = e^x$ or $e^y \cdot e^{e^y} = e^{e^x}$ or $(x-y) + \ln(x-y) = x$
- PAK112 – General solution for $y \cdot \ln(y) = x$ or $\ln(y^y) = x$ or $y^y = e^x$ <-New
- PAK112 – General solution for $(1/y)\ln(y) = x$ or $\ln(y^{1/y}) = x$ or $y^{1/y} = e^x$
- PAK113 – Solution for $(ay^2 + by + c)e^{y(1+dy)} = e^x$ or $y(1+dy) + \ln(ay^2 + by + c) = x$
- PAK114 – Derivatives $dW(e^x)/dx$ and $d^2W(e^x)/dx^2$
- PAK115 – Indefinite integrals $\int W(x)dx$ and $\int W(e^x)dx$ <-New
- PAK116 – Adding two $W(e^x)$ terms to a single $W(e^x)$ term <-New
- PAK117 – Subtracting two $W(e^x)$ terms to a single $W(x)$ term <-New
- PAK118 – W-function for voltage gain and distortion HD2 & HD3

Level 2 – How to solve multi-diode circuits

- PAK201 – Multi-diode solution for 2D+R
- PAK202 – Multi-diode solution for $R + [D/(D+D)]$
- PAK203 – Multi-diode solution for $D + (R/(D+R))$
- PAK204 – Multi-diode solution for $(V+D+R)/(-V+D+R)$ for Basic PP CE
- PAK205 – Multi-diode solution for $R + (V+D)/(-V+D)$ for I_{in} & V_{in}
- PAK206 – Multi-diode solution for $R + (V+D+R)/(-V+D+R)$ for Basic PP CC Follower
- PAK207 – Multi-diode BJT modelling for V_{ce} saturation
- PAK208 – Multi-diode BJT modelling for Beta-fall
- PAK209 – Multi-diode BJT modelling for Early effect
- PAK210 – Multi-diode BJT modelling for Thermal effects
- PAK211 – Multi-diode BJT modelling for Electro-thermal effects
- PAK212 – MOSFET modelling
- PAK213 – Basic Triode Tube modelling

Level 3 – Multi-transistor circuit

- PAK301/214 – Basic Darlington pair
- PAK302/215 – Basic Complementary Pair (CFP)
- PAK303/216 – Basic Voltage Feedback Pair
- PAK304/217 – Basic Differential pair (BJT LTP)
- PAK305/218 – Basic BJT Current Mirror
- PAK306/218a – Basic Current Source with feedback

PAK307/218b – The Wilson Mirror
PAK308/219 – Basic V_{be} multiplier
PAK309/220 – Basic Cascode pair
PAK310/220 – Basic Folded Cascode
PAK311/221 – Rush Pair or Composite Pair
PAK312/304 – Current Feedback Amplifier CFA as a CFP

Level 4 – Push-Pull amplifier circuits

PAK401/301 – Basic CE Push-Pull [copy of PAK204+ more]
PAK402 – Basic CC Push-Pull Follower [copy of PAK206+ more]
PAK403/302 – Darlington push-pull
PAK404/303 – CFP Push-Pull
PAK405/402 – Basic Lin topology DC solution
PAK406/403 – Lin topology with Beta-fall, Early effect & Saturation DC solution
PAK407/404 – Lin topology with Thermal effects
PAK408/405 – Basic 3 stage topology DC solution
PAK409/406+407 – Current Driven Class-B splitting DC solution
PAK410 – Current Source Driven PP CE (LA Vol.13) DC solution
PAK411 – DoubleCross™ (Class-AB+C) CE
PAK412/307 – DoubleCross™ (Class-AB+C) CC Follower
PAK413/408 – Faran-AB current driven DoubleCross™ DC solution
PAK414/409 – Cube-law amplifier general DC solution

Level 5 – Further research

PAK501 – New FFT Analytic Harmonic Analysis (AHA) using derivatives of general W solutions
PAK502 – DDE W-stability analysis (ref [Ulsoy](#)). Can it be used for approximate AC analysis?
PAK503 – Software manipulation of transistor circuit analytic W-solutions using MathCAD etc.

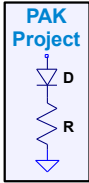
PAK600 – References, supplementary material

PAK601 – References for all PAK material
PAK602 – Supplementary material

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No liability is accepted. Use the "Contact" section to email me.

Cheers, Ian Hegglun ~60 Sections

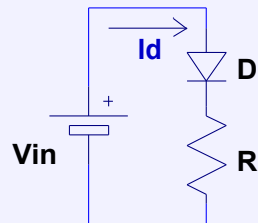
LOGO & BADGES



click image, linked to website

PAK Project 102

How to solve for $I_d = ?$



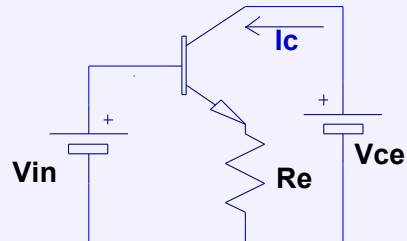
Easy ...

Use Lambert's W-function

$$I_d = V_t / R * W[e^{(V_{in} - V_k) / V_t}]$$

PAK Project 103

How to solve for $I_c = ?$



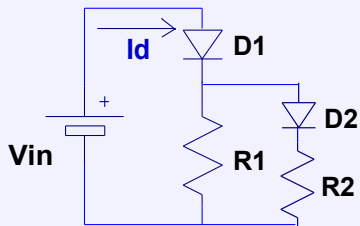
Easy ...

Use Lambert's W-function

$$I_c = V_t / R_e * W[e^{(V_{in} - V_k) / V_t}]$$

PAK Project 202

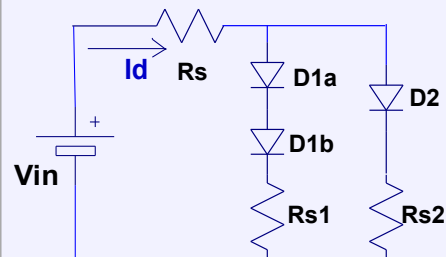
How to solve for $I_d = ?$



Use Lambert's W-function

PAK Project 203

How to solve for $I_d = ?$



Use Lambert's W-function